

PRITHVEX Methodology

Data, Modelling & Simulation Pipeline · Team Envirex

1. Purpose of this Document

This document describes the end-to-end methodology used to build, train, and operate the PRITHVEX model that powers the ENVIREX risk-intelligence dashboard. It covers data sourcing, feature engineering, the modelling approach, the simulation layer, and the operational feedback loop that lets the system keep improving after deployment.

2. Data Ingestion & Integration

PRITHVEX is designed to ingest structured and unstructured signals from multiple source types simultaneously:

- ERP and procurement systems, parsed via structured and NLP pipelines for unstructured text.
- IoT sensors and industrial monitoring feeds for live operational telemetry.
- Satellite intelligence feeds for environmental and infrastructure conditions.
- Weather APIs, ESG reports, and real-time geopolitical disruption feeds.
- Knowledge graphs that encode supplier – disruption – logistics relationships across the network.

All sources are onboarded through a common ingestion layer before being passed downstream, so that every model in the ensemble sees a consistent, time-aligned feature set.

3. Prediction Engine

The prediction engine combines four model families, each chosen for a complementary strength:

Model	Role
XGBoost / LightGBM	Tabular feature interactions; battle-tested in Tier-1 enterprise SCM
LSTM networks	Temporal demand pattern recognition
Transformer models	Anomaly detection and logistics disruption forecasting
Ensemble combiner	Blends model outputs into a single calibrated point estimate per SKU/node

Outputs are produced across three forecast horizons — 7-day, 30-day, and 90-day — so that both tactical and strategic procurement decisions can be supported from the same underlying model.

4. Uncertainty Calibration Methodology

Raw ensemble outputs are passed through a MAPIE-based split conformal prediction layer to produce confidence-bounded intervals per supplier, route, and warehouse node. This step is what separates PRITHVEX from a conventional point-forecast model: every prediction is shipped with a statistically grounded interval rather than a bare number. The full calibration procedure, including the Expected Calibration Error (ECE) check used to decide whether a model needs to be retuned, is detailed in the companion calibration report.

5. Network Risk Intelligence Layer

On top of single-node forecasts, PRITHVEX builds a Tier-1/2/3 supplier dependency graph and runs Monte Carlo shock propagation across it. This lets the system estimate how a localised disruption (for example a single supplier shutdown) is likely to cascade through downstream tiers before it actually reaches an operational stage, surfacing hidden bottlenecks and procurement vulnerabilities ahead of time.

6. Scenario Simulation Workflow

The scenario simulation panel lets a user select variables and risk parameters and trigger the Monte Carlo engine on demand. The engine runs thousands of probabilistic scenarios and groups the resulting outcome paths into three branches:

- Best-case branch — low risk outcome path.
- Base-case branch — expected risk trajectory.
- Worst-case branch — high disruption scenario.

Each branch is paired with an automated, risk-aware recommendation rather than a bare probability number, so the output is directly actionable for a procurement or risk team.

7. Explainable Recommendation Engine

Every recommendation is backed by a SHAP attribution trail. A representative attribution snapshot from a live scenario is shown below.

Driver	Contribution	Risk Category
Heavy Rainfall Index	+0.42	Environmental Risk
Typhoon Probability	+0.34	Climate Risk
Port Congestion (Vietnam)	+0.28	Logistics Risk
Supplier Shutdown Probability	+0.21	Supply Risk
Lead-Time Variance	+0.12	Operational Risk

8. Decision Implementation & Feedback Loop

When a user acts on an alert or recommendation, the action and its outcome are logged back into the system. This closes the loop: the model is periodically recalibrated against realised outcomes, which is what allows PRITHVEX to describe itself as an adaptive, self-improving model rather than a static one-off forecast.

9. Underlying System Layers

- Cloud infrastructure — AWS / Azure / GCP / Kubernetes for elastic deployment.
- Bayesian core — probabilistic inference and uncertainty quantification.
- Simulation engine — Monte Carlo scenario planning shared across all dashboard panels.